

WAX MELTER

AI-powered sports card market intelligence built to find mispriced inventory before the market does.

CORE QUESTION: Where is the sports card market wrong right now?

The Product

Wax Melter is a decision engine for sports card resellers. It scans marketplace inventory, identifies the exact card, estimates fair market value, measures liquidity and player momentum, calculates a profitable maximum purchase price, and ranks each opportunity as **BUY, WATCH, or PASS**.

1. MARKET SCANNER Monitor auctions and Buy It Now listings. Surface underpriced cards, auctions ending soon, weak titles, misspellings and low-visibility listings.	2. EXACT CARD ID Normalize player, year, set, card number, parallel, serial number, rookie/1st status, grading company and grade.
3. COMP ENGINE Weight exact recent sales, remove outliers and calculate confidence-weighted fair market value.	4. PROFIT ENGINE Estimate resale proceeds after fees/shipping. Output max buy, expected profit and expected ROI.
5. OPPORTUNITY SCORE Rank discount to FMV, liquidity, card hierarchy, player momentum, seller risk, grading upside and catalysts.	6. SELLING ENGINE Generate listings and pricing, track inventory and eventually automate repricing.

Example Decision Output

Listing	Current	FMV	Max Buy	ROI	Score	Action
Flagship rookie parallel - PSA 10	\$42	\$71	\$49	31%	91/100	BUY

V1 principle: Wax Melter recommends. The user approves. Automated purchasing comes only after the model proves its pricing and profitability assumptions against real outcomes.

WAX MELTER - DEVELOPER BUILD BRIEF

Build the intelligence layer first. Automate transactions only after the decision engine is validated.

V1 Scope

Market	eBay-first
Sports	NFL + MLB initially; NBA after validation
Card universe	Flagship rookies, 1st Bowman Chrome, Prizm/Optic/Topps Chrome, numbered parallels, PSA 9/10
Price band	\$10-\$100 acquisition targets
Output	BUY / WATCH / PASS + FMV + max buy + expected profit/ROI + confidence
Execution	Human approval only - no autonomous buying in V1

Recommended Architecture

INGESTION	eBay APIs / approved market-data sources -> live listings, item details and historical sales where available.
NORMALIZATION	LLM-assisted parser + structured checklist data -> canonical card identity.
VALUATION	Weighted median comps, recency/exact-match weighting, outlier removal, trend and liquidity adjustments.
SCORING	Discount to FMV + liquidity + player momentum + card hierarchy + seller risk + grading upside + catalysts.
DATA	SQLite first; migrate to PostgreSQL. Store cards, listings, comps, decisions, purchases, inventory and sales.
UI	Streamlit V1 dashboard; React/Next.js later if productized.

Core Data Model

Card: player, sport, year, manufacturer, set, card_number, parallel, serial_limit, rookie_status, first_bowman, autograph, grade_company, grade.

Listing: marketplace_id, seller, price, shipping, auction_end, title, images, canonical_card_id. **Comp:** sold_price, sold_date, match_quality, sale_format. **Decision:** FMV, confidence, liquidity_score, opportunity_score, max_buy, expected_net, expected_profit, expected_ROI, action.

Phased Build

01	PROTOTYPE Manual card query -> exact identity -> comp set -> FMV -> max buy -> BUY/PASS.
02	SCANNER Scan the defined card universe and rank live opportunities; alert on high-confidence deals.
03	PORTFOLIO Track actual purchases/sales vs predictions. Add inventory and repricing.
04	AUTOMATION After validation, add approved bidding/buying and automated listing workflows.

Success Criteria Before Autonomous Buying

Run at least **100-200 recommendations** through paper trading / human approval. Measure FMV prediction error, realized ROI, sell-through time, false card matches and score calibration. Initial target: **20%+ realized ROI after all costs**, with strict bankroll and per-player exposure limits.

THE PRODUCT: A system that does not merely price cards - it continuously identifies where market price and probable resale value diverge.